

Leo FLACK

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EDUCATION

University of Cambridge <i>Part III of the Mathematical Tripos (MASt in Mathematics)</i> Selected coursework: <i>Advanced Probability, Stochastic Calculus, Information Theory, Random Walks & Phase Transitions.</i>	Oct 2026 – Jun 2027 Cambridge, UK
Ecole Polytechnique <i>Cycle Ingenieur Polytechnicien – Mathematics & Physics</i> GPA: 3.97/4.0. Coursework: <i>Martingales & Markov Chains, Optimization & Control, Statistical Physics, Econophysics.</i>	Sep 2023 – Aug 2026 Palaiseau, France
Lycee Louis-le-Grand <i>Classes Préparatoires aux Grandes Ecoles (CPGE) – Mathematics & Physics (MP*)</i>	Sep 2021 – Jun 2023 Paris, France

HONORS & AWARDS

National University Entrance Exams (2023): ranked **2nd** / 5,000+ (Mines Paris), **6th** (Ecole Polytechnique), **9th** (ENS Ulm)
Carnot Scholarship (2026): merit-based award, **1 recipient in cohort**, Ecole Polytechnique
Saven European Scholarship (2026): **£25,000 award** (Oxford physics, declined for Cambridge math)
Military Decorations (2024): NATO Vigilance Medal; Bronze Medal of National Defense (France) for exceptional service

RESEARCH EXPERIENCE

Finite Size Effects and Extreme Statistics in Hopfield Neural Networks – UChicago <i>Visiting Researcher – Advisor: Prof. Vincenzo Vitelli</i> First-author paper in preparation. Demonstrated numerically that finite delayed non-reciprocal Hopfield networks differ qualitatively from their infinite-dimensional limit. - Using Python for sample-resolved continuation, delay dynamics, spectral stability and Lyapunov analysis, uncovered how an ensemble of local bifurcations give birth to a phase transition from static to sequential memory recall.	Mar 2026 – Aug 2026 Chicago, IL, USA
High-Dimensional Learning and Statistical Physics of Autoencoders – ENS Paris <i>Research Project – Advisor: Prof. Remi Monasson</i> - Studied how data geometry controls autoencoder interpolation. Showed that out-of-sample error is governed by covariance eigenvalue decay using synthetic high-dimensional experiments - Derived via the Replica method the training error of an asymmetric autoencoder as an explicit function of the sample-to-dimension ratio and the data covariance spectrum, separating under- and over-parametrized regimes.	Sep 2025 – Mar 2026 Paris, France
Plasma Modeling for Air-Breathing Electric Propulsion – Ecole Polytechnique <i>Group research project – Advisor: Dr. Federico Petronio</i> - Co-authored the peer-reviewed publication “Global plasma modeling of an air-breathing gridded ion thruster for very low Earth orbit applications” (2026), <i>Physics of Plasmas</i>, DOI: 10.1063/5.0327675 - Established a feasibility window for air-breathing electric propulsion by developing a 9-species plasma ODE solver and mapping atmospheric conditions, discharge power and losses to the thrust–drag balance.	Sep 2024 – Sep 2025 Palaiseau, France
Complex Amoebas and Tropical Geometry – Ecole Polytechnique <i>Mathematical Essay – Advisor: Prof. Omid Amini</i> Proved Hausdorff convergence of rescaled amoebas to tropical curves and the algebraic–tropical zero set correspondence.	Sep 2024 – Jan 2025 Palaiseau, France

ADDITIONAL EXPERIENCE

Thales Japan K.K. <i>Internship as Innovation & Emerging Technologies Analyst</i> Evaluated 15+ deep-tech startups using technical metrics and authored a report on Japan’s quantum-computing ecosystem.	Jun 2025 – Sep 2025 Tokyo, Japan
French Navy – Frigate Aquitaine <i>Military service as Officer Cadet</i> Appointed Officer of the Watch during a 12-week NATO deployment in the Baltic Sea: led a team of 12 sailors.	Nov 2023 – Apr 2024 Brest, France

SKILLS

Programming: Python (NumPy, SciPy, PyTorch, Matplotlib), OCaml ; Scientific computing (ODE integration, spectral analysis).
Languages: French (native), English (IELTS 8.5/9), Italian (advanced)

LEADERSHIP & INTERESTS

Teaching: Mathematics & physics oral examiner for CPGE students
Service: Raised €16k for education projects in Senegal; managed €40k in microloans to 120+ families in Guatemala
Interests: Competitive swimming, Model UN (Best Speaker, Geneva), scouting (10+ years), chess, poker